



# Komal's Role: LLM Researcher

## SCROPIDS PROJECT

Komal serves as the **LLM Researcher** for the ScropIDS project — an AI-powered intrusion detection system that uses a Large Language Model to analyse aggregated security events and return structured threat analysis. Her responsibility was to evaluate the landscape of available LLM options, compare models across cost, API compatibility, and cybersecurity suitability, and recommend an integration strategy that would make ScropIDS both flexible and intelligent.

### Model Comparison

Evaluated cloud, local, and hybrid LLM options for suitability

### Cost Analysis

Identified free and low-cost access paths for capstone constraints

### Integration Strategy

Mapped API compatibility with ScropIDS backend architecture

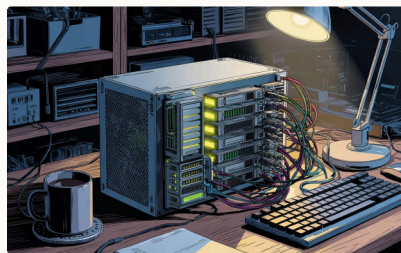
# Research on Available LLM Options

Komal conducted a comprehensive survey of the LLM ecosystem, evaluating both cloud-hosted and locally deployable models to determine which would best serve ScropIDS's need for real-time cybersecurity log analysis.



## Cloud-Based Models

Assessed major providers including OpenAI, Anthropic, and Google. Evaluated response quality, rate limits, and suitability for structured JSON output in security contexts.



## Local Models via Ollama

Explored self-hosted options such as Mistral and LLaMA running through Ollama — enabling offline operation, data privacy, and zero API cost for development and testing.



## API Compatibility & Reliability

Benchmarked API access patterns, OpenAI-compatible endpoints, uptime reliability, and the quality of responses when processing structured security event payloads.

## KEY RECOMMENDATION

# OpenRouter: Komal's Strategic Recommendation

After evaluating multiple integration paths, Komal recommended **OpenRouter** as the primary LLM gateway for ScropIDS. OpenRouter provides a single, unified OpenAI-compatible API endpoint that routes requests to dozens of different models — dramatically simplifying the backend architecture.

### Unified API Gateway

One endpoint, many models. ScropIDS can switch between GPT-4o, Claude, Mistral, or others without touching the backend integration code — only a model identifier changes.

### Free & Low-Cost Access

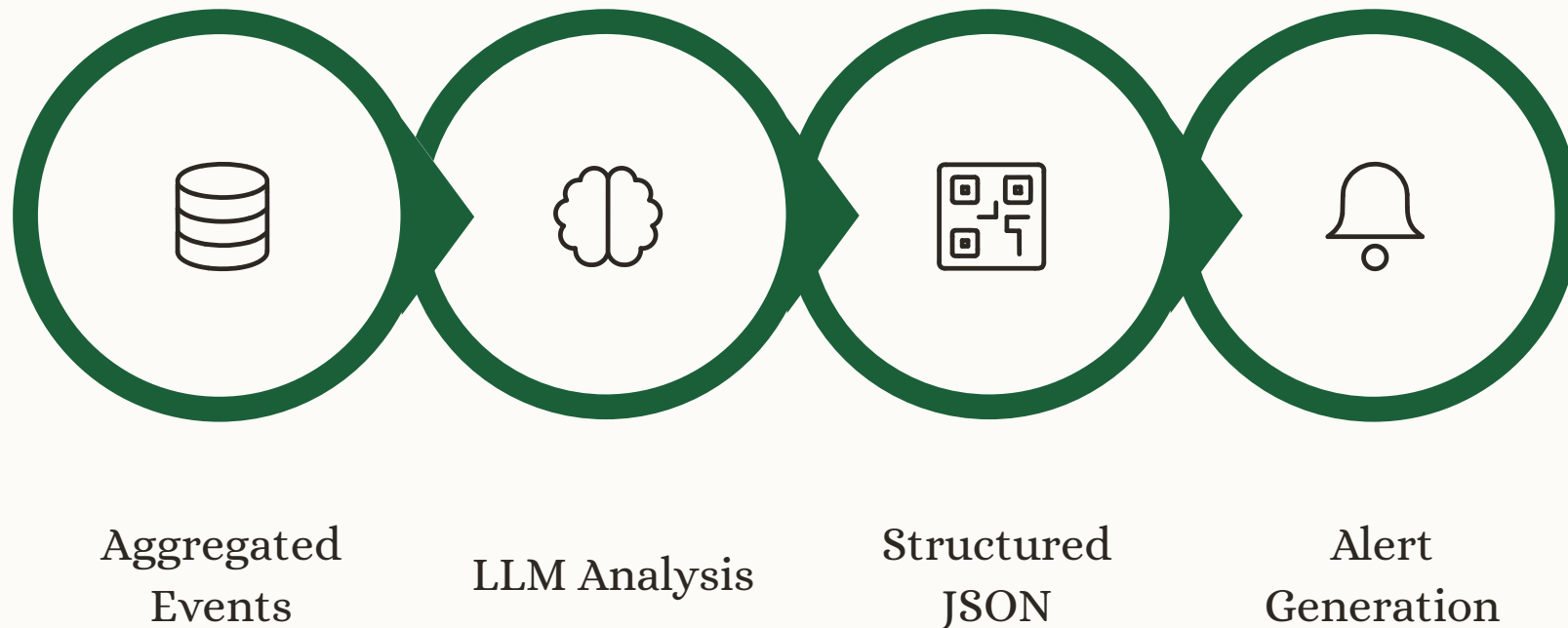
Several models on OpenRouter offer free or heavily discounted tiers — ideal for a capstone project where cost control and iterative testing are essential priorities.

### Rapid Model Experimentation

The team can benchmark response quality across multiple models for security analysis tasks, choosing the best performer without architectural rework or downtime.

# LLM Integration in ScropIDS

Komal's research directly shaped how the LLM is embedded within the ScropIDS pipeline. The integration follows a structured flow — from raw event aggregation through AI analysis to actionable security alerts.



The backend validates every LLM response against a strict schema before alert creation, ensuring that only well-formed, threshold-exceeding findings surface as actionable incidents. This design keeps the system both intelligent and operationally reliable.

## Supported Backends

- OpenAI-compatible APIs
- OpenRouter multi-model gateway
- Local Ollama deployments

## Output Contract

- Structured JSON with severity score
- Threat category classification
- Human-readable reasoning summary



## FINAL CONTRIBUTION

# Komal's Impact on ScropIDS

Komal's thorough LLM research was the intellectual foundation that gave ScropIDS its defining capability — **intelligent, AI-powered threat analysis**. By mapping the model landscape, identifying cost-effective access strategies, and championing OpenRouter's unified API approach, she ensured the project remained flexible, testable, and production-ready.

### Flexible Architecture

Any OpenAI-compatible model can be swapped in without backend changes, future-proofing the system.

### Cost-Effective Design

Free and low-cost model tiers identified through OpenRouter kept the capstone within budget constraints.

### AI-Powered Detection

LLM integration elevated ScropIDS from a rule-based tool to an intelligent, reasoning-capable threat analyser.

"The LLM research gave ScropIDS its intelligent threat analysis capability."